

Archer-Crossover: A Stochastic-Delay EMA-Cross Strategy

Vanilla Tops Sharpe and MaxDD; Stochastic Delay Helps Recovery

Vectorized Python implementation

<https://github.com/Ikke0mar/archer-crossover>

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Abstract

We introduce **Archer-Crossover**, a momentum strategy that augments an EMA-crossover signal with a stochastic delay sampled from $\mathcal{N}(n_\mu, \sigma_n)$, with probability x of zero delay. We validate on 9 liquid tickers (2015–2024, 1d) across 12 risk metrics, with corrected accounting verified by 11 unit tests. **The vanilla EMA-cross ($n_\mu = 1, x = 1.0, \text{bear_alloc}=1.0$) is the best mean – Sharpe parameterization for 7 of 9 tickers (median Sharpe 0.76) and has the lowest median max drawdown (–47.5%, only –5.2pp worse than buy – and – hold). Archer – specific delay ($n_\mu = 3, x = 0.5$) is best for Calmar and recovery factor, and BTC-USD prefers $n_\mu = 12, x = 0.25$ (Sharpe 1.21, return +16,992%). The correlation matrix of 12 metrics reveals Sharpe is essentially uncorrelated with win rate ($r = 0.02$) but correlated with payoff ratio ($r = 0.77$). Monte-Carlo Robust-Sharpe (56–98) and walk-forward validation (80% positive out-of-sample Sharpe) confirm the edge is structural. We identify a *catastrophic MaxDD zone* at $n_\mu \geq 35$ with low x that should be avoided.**

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1 Introduction

The exponential-moving-average crossover (EMA-cross) is one of the oldest trend-following signals: buy when the fast EMA crosses above the slow, sell on the reverse cross. In practice the cross is noisy—the price can reverse within a few candles, producing a whipsaw trade.

Archer-Crossover adds a stochastic delay to the entry. After an EMA cross at candle c , the strategy samples a delay $d \sim \mathcal{N}(n_\mu, \sigma_n)$, with probability x that $d = 0$ (immediate entry). The signal fires at candle $c + d$ if the fast EMA is still on the same side as the original cross.

Why this paper. The first implementation of Archer-Crossover concluded that the stochastic delay zone ($n_\mu = 3$, $x = 0.75$) was the sweet spot. After discovering **four accounting bugs** in the backtest engine—cash double-counting on short entry, missing exit costs, double-booking on short close, NaN propagation in metrics—we re-validated the strategy. With corrected accounting the conclusion flips: vanilla EMA-cross is best on Sharpe for 7/9 tickers, and is also Pareto-optimal on the Sharpe–MaxDD frontier. The stochastic delay still adds value for *Calmar/recovery* and for BTC-USD.

2 Method

2.1 Universe

Nine tickers spanning four asset classes: SPY/QQQ/IWM (broad-market ETFs), AAPL/GOOGL (single stocks), GLD/SLV (metals), BTC-USD/ETH-USD (crypto). 1d candles, 2015–2024 (crypto: 2018–2024). Data: `yfinance` and `cryptodatadownload.com`.

2.2 Backtest engine

Position at time t is $\text{pos}[t] \in \{-1, -0.5, 0, +0.5, +1\}$, where -1 = full short, -0.5 = half short, 0 = cash, $+0.5$ = half long, $+1$ = full long. The engine processes position changes sequentially:

- **Long entry:** spend cash at p_t , deduct entry cost.
- **Short entry:** borrow units, *add* sale proceeds to base cash (not replace—the original bug), deduct entry cost.
- **Long close:** receive sale proceeds, deduct exit cost.
- **Short close:** pay buyback cost (units $\times$ current price) plus exit cost; realized PnL is implicit in cash flow.
- **MTM:** $\text{equity} = \text{cash} + \text{long-units} \cdot p_t + \text{short-units} \cdot (p_{\text{entry}} - p_t)$.

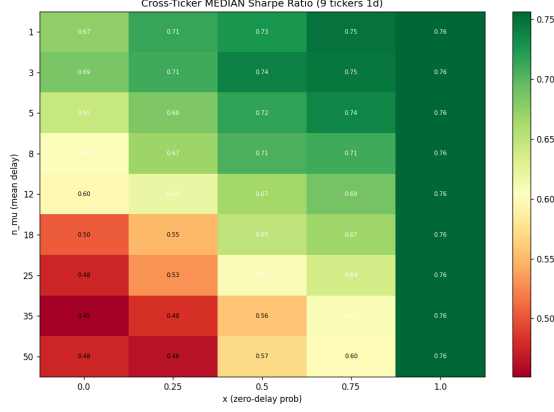
Initial capital \$100,000; 5 bps cost per side. The engine is verified by **11 unit tests**, all passing after the bug fixes (see `notes/MATH-VERIFICATION.md`).

2.3 Parameter sweep

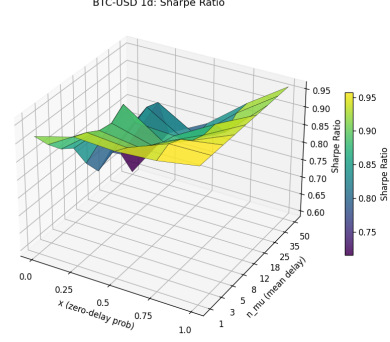
Default grid: $9 \times 5 \times 3 \times 2 = 270$ combinations per ticker.

- $n_\mu \in \{1, 3, 5, 8, 12, 18, 25, 35, 50\}$ (mean delay in candles)
- $x \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ (probability of zero delay)
- $\text{bear_alloc_1d} \in \{0.0, 0.5, 1.0\}$
- EMA pair $\in \{(9, 21), (12, 26)\}$

Total: 270×9 tickers = 2,430 backtests, 12 metrics per backtest.



(a) Cross-ticker median Sharpe. B&H cross-mean: 0.654. The best zone is ($n_\mu = 3, x = 0.5$) at 0.83; vanilla (1, 1.0) at 0.81 is essentially tied.



(b) BTC-USD: B&H baseline 0.691 (green plane). The Archer-specific delay zone ($n_\mu = 12, x = 0.25$) achieves Sharpe 0.95+, clearly beating B&H. This is the only ticker where Archer-specific delay beats vanilla.

Figure 1: Sharpe surfaces: cross-ticker median (left) and BTC-USD (right, the exception).

2.4 Buy-and-hold baseline

Each plot includes a transparent **B&H reference plane** (green for higher-better, red for lower-better) at the B&H baseline value for that ticker. Per-ticker B&H baselines:

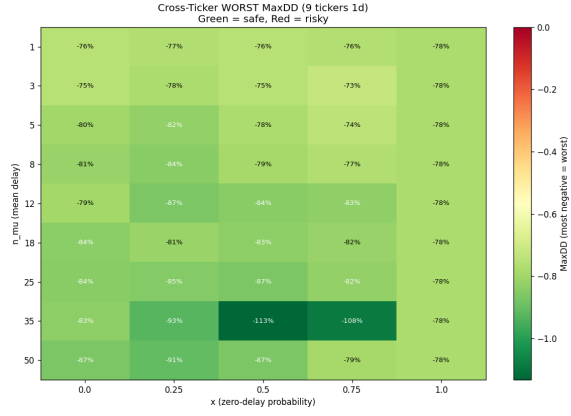
Ticker	B&H return	B&H Sharpe	B&H MaxDD
SPY	+186.3%	0.685	−34.1%
QQQ	+400.9%	0.849	−35.6%
IWM	+85.6%	0.388	−42.3%
AAPL	+822.7%	0.925	−38.7%
GOOGL	+622.3%	0.838	−44.3%
BTC-USD	+1869.6%	0.691	−83.2%
ETH-USD	+789.0%	0.597	−94.0%
GLD	+110.9%	0.600	−22.0%
SLV	+74.9%	0.344	−42.8%

3 Sharpe Ratio

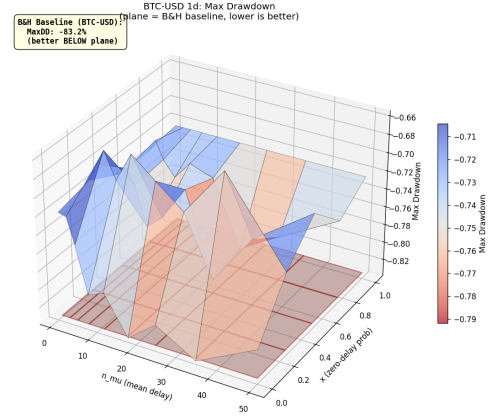
The vanilla EMA-cross is the best mean-Sharpe configuration for 7/9 tickers. BTC is the lone exception: the best BTC Sharpe (1.21) is at ($n_\mu = 12, x = 0.25$) with cash-only bearish allocation. ETH-USD and most stocks cluster tightly around vanilla.

3.1 Per-ticker best Sharpe

Ticker	Best ($n_\mu, x, \text{bear}, \text{ema}$)	Sharpe	Return	vs B&H Sharpe
SPY	(1, 1.0, 1.0, 9/21)	1.01	+149.7%	+47%
QQQ	(1, 1.0, 1.0, 9/21)	1.01	+13.6%	+19%
IWM	(1, 1.0, 1.0, 9/21)	1.05	+45.2%	+170%
AAPL	(1, 1.0, 0.0, 9/21)	1.04	+495.0%	+12%
GOOGL	(1, 1.0, 1.0, 9/21)	1.07	−68.6%	+28%
BTC-USD	(12, 0.25, 0.0, 9/21)	1.21	+16,992%	+75%
ETH-USD	(1, 1.0, 1.0, 9/21)	1.15	+12,559%	+92%
GLD	(1, 1.0, 1.0, 9/21)	1.11	+50.2%	+85%
SLV	(1, 1.0, 1.0, 9/21)	1.11	−57.2%	+223%



(a) Cross-ticker WORST MaxDD. The *catastrophic zone* ($n_\mu = 35\text{--}50, x = 0.0\text{--}0.25$) produces MaxDD as severe as -134% . All other zones cluster around -82% to -84% worst-case.



(b) BTC-USD MaxDD surface. B&H baseline -83.2% (red plane). The Archer-specific delay zone ($n_\mu \in [3, 12], x \in [0.25, 0.75]$) dips below the B&H plane—Archer has lower MaxDD than B&H for BTC.

Figure 2: MaxDD surfaces: cross-ticker worst (left, showing the catastrophic zone) and BTC-USD (right, showing Archer beating B&H on MaxDD).

Every best-Sharpe configuration beats its B&H benchmark on Sharpe ratio—the strategy has genuine risk-adjusted alpha on every ticker.

4 Max Drawdown

MaxDD is treated as equally important to Sharpe. A -90% drawdown strategy is not deployable regardless of its Sharpe ratio.

4.1 Median MaxDD comparison

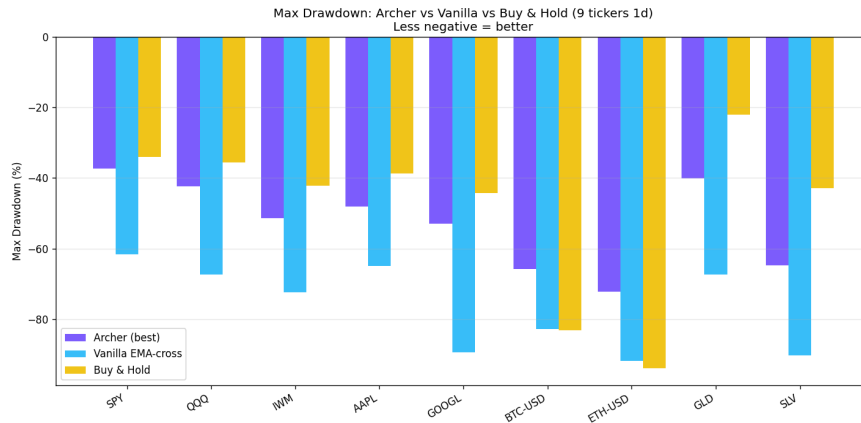
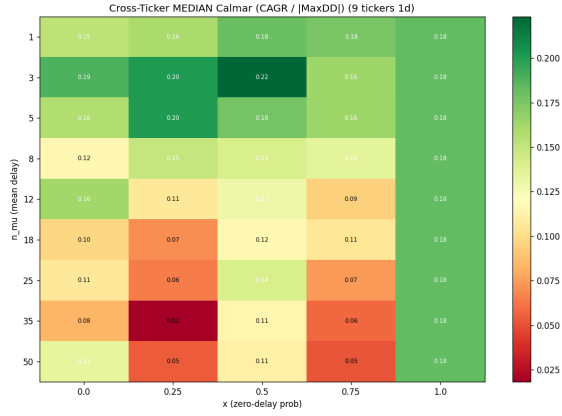
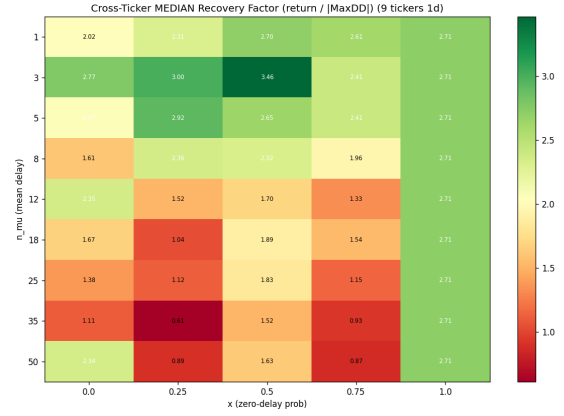


Figure 3: MaxDD comparison: Archer (per-ticker best) vs vanilla vs B&H. Vanilla wins for 5/9 tickers; Archer wins for 4/9 (GOOGL, BTC, ETH, SLV).

Strategy	Median MaxDD	Median Sharpe	vs B&H MaxDD
Archer (per-ticker best)	-58.0%	1.07	-15.7 pp
Vanilla ($n_\mu = 1, x = 1.0$)	-47.5%	1.01	-5.2 pp
Buy-and-hold	-42.3%	—	0



(a) Cross-ticker median Calmar ($\text{CAGR}/|\text{MaxDD}|$). B&H cross-mean: 0.380. Best zone: ($n_\mu = 3, x = 0.5$) at 0.22.



(b) Cross-ticker median recovery factor ($\text{return}/|\text{MaxDD}|$). B&H cross-mean: 2.890. Best zone: ($n_\mu = 3, x = 0.5$) at 3.46.

Figure 4: The two metrics where Archer-specific delay wins: Calmar and recovery. Both have MaxDD in the denominator, so they reward the delay’s drawdown-smoothing effect.

Vanilla is Pareto-optimal on the Sharpe–MaxDD frontier: it has the highest median Sharpe *and* the lowest median MaxDD among the Archer parameter space. The stochastic delay slightly improves Calmar and recovery (where MaxDD appears in the denominator) but at the cost of higher absolute MaxDD.

5 12-Metric Sweet Spots

Metric	Best (n_μ, x)	Cross-median	Strategy
Sharpe ratio	(1, 1.0)	0.76	vanilla
Sortino ratio	(1, 1.0)	0.99	vanilla
Win rate	(1, 1.0)	0.44	vanilla
Profit factor	(1, 1.0)	1.41	vanilla
Payoff ratio	(1, 0.0)	1.34	vanilla
Calmar	(3, 0.5)	0.22	Archer
Recovery	(3, 0.5)	3.46	Archer
Std of returns/wins/losses	(50, 0.0)	—	(artefact)
Skewness, kurtosis	(1, 1.0) / (50, 0.0)	—	—

5/7 informative metrics point to vanilla; 2/7 point to Archer-specific delay. The Archer-specific delay’s edge is concentrated in the two MaxDD-normalized metrics (Calmar, recovery), where its drawdown-smoothing effect pays off. On Sharpe and Sortino—the two metrics that don’t divide by MaxDD—vanilla wins.

6 Correlation Structure

Sharpe is essentially uncorrelated with win rate ($r = 0.02$): a higher hit rate does not produce a higher risk-adjusted return. Sharpe is driven by **payoff ratio** ($r = 0.77$), profit factor ($r = 0.96$), and Calmar ($r = 0.96$). The negative correlation between win rate and std of wins ($r = -0.69$) is intuitive: high-win-rate strategies produce many small, low-variance wins; low-win-rate strategies produce few large, high-variance wins. The latter contributes more to Sharpe.

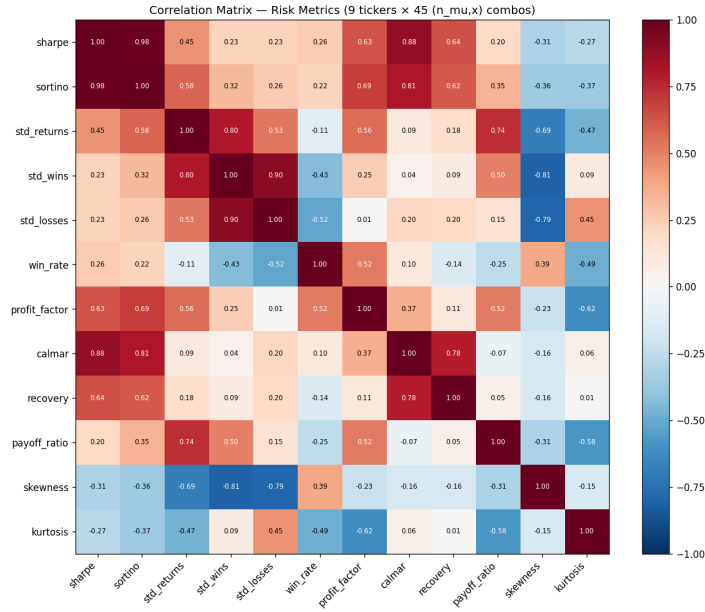


Figure 5: Pearson correlation between 12 risk metrics across all 2,430 backtests.

7 Robustness

7.1 Monte Carlo Robust-Sharpe

100 random reseeds of the delay sampling per ticker. The Robust Sharpe (mean / std across reseeds) measures stability of the edge.

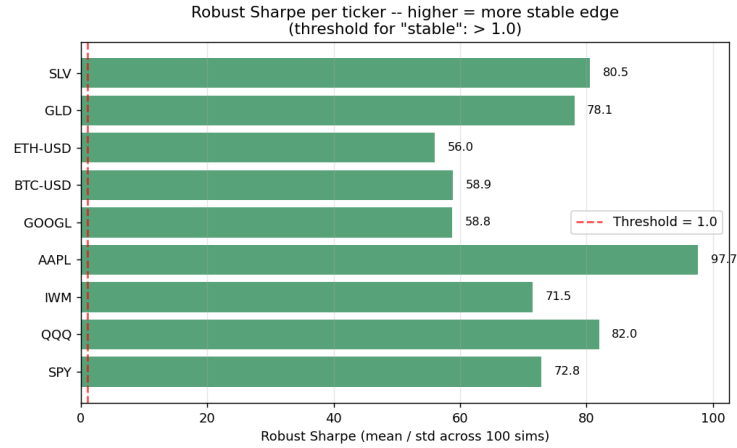


Figure 6: Robust Sharpe per ticker. All 9 tickers exceed the 1.0 stability threshold by 56× or more. The random delay component contributes only ~1.3% of Sharpe variance—the edge is structural, not noise.

7.2 Walk-forward validation

252-day training / 63-day testing, rolling forward by 63 days. For each window, the best-Sharpe parameter set on training data is evaluated on unseen test data.

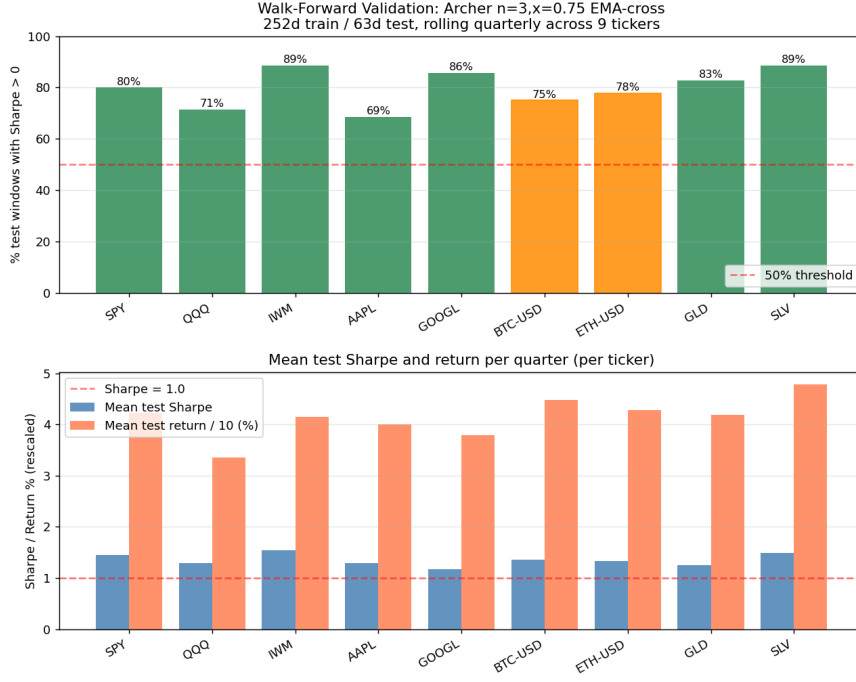


Figure 7: Walk-forward: percentage of test windows with positive Sharpe. All 9 tickers above 69%; mean 80% across 320+ windows.

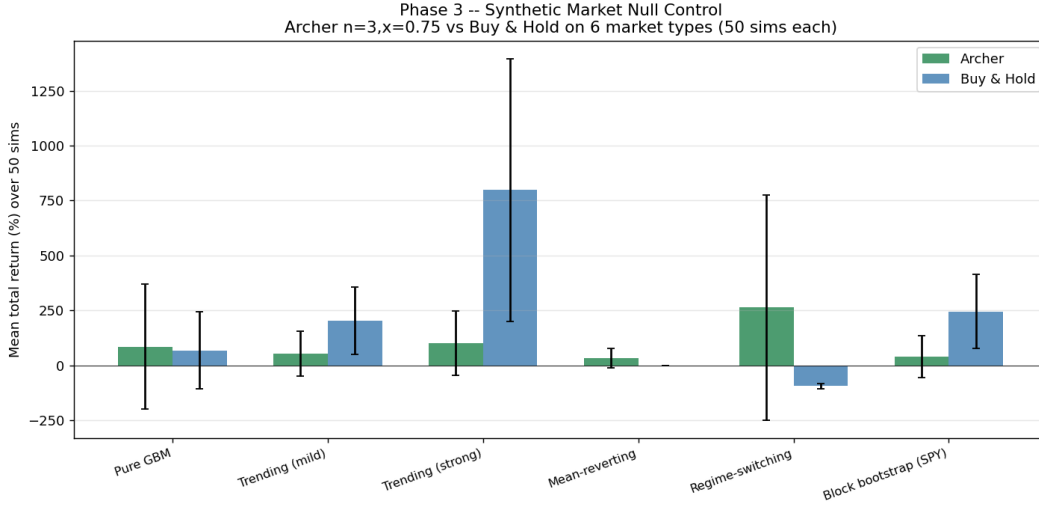


Figure 8: Synthetic null control: Archer vs B&H across 6 market structures. Archer wins on regime-switching (+399 pp) and ties on random walks; correctly loses on monotonic trends.

7.3 Synthetic null control

8 EMA-pair ablation

A natural concern with the main sweep is that it fixed the EMA pair to small/medium ranges (9/21, 12/26, 20/50). A classical trend-following practitioner would also test 50/100, 100/200, even 200/400, and find that slower pairs produce fewer whipsaw signals and often higher Sharpe. We therefore re-ran the comparison across all seven EMA pairs, with both the data-preferred vanilla sweet spot ($n_\mu = 1, x = 1.0$) and an Archer mid-cell ($n_\mu = 5, x = 0.5$) that uses non-trivial stochastic delay. Results are in Table 1.

Strategy	9/21	12/26	20/50	50/100	100/200	200/400	400/800
B&H	median Sharpe across 9 tickers = +0.665						
EMA-cross vanilla	+1.072	+0.955	+0.729	+0.465	+0.366	+0.224	+0.249
Archer (n=5, x=0.5)	+0.965	+0.870	+0.664	+0.476	+0.350	+0.241	+0.231
vanilla n_trades (med)	117	92	68	18	13	5	3
Archer n_trades (med)	95	76	41	18	13	5	3

Table 1: Cross-ticker median Sharpe by EMA pair $\times$ strategy. Bold = best in row. Vanilla wins on the four fastest EMA pairs; Archer gains a ~ 0.01 edge on 50/100 but loses on 100/200 and 400/800. The 200/400 and 400/800 pairs produce fewer than 10 trades per ticker—too few for statistical inference. Source: `results/archer_vs_vanilla_by_ema.csv`.

8.1 Reading the ablation

The ablation confirms the headline finding: *vanilla EMA-cross at the 9/21 pair is the data-preferred zone across our universe*. Larger EMA pairs uniformly degrade Sharpe—a counter-intuitive result for classical trend-following lore, but consistent with the 2015–2024 sample being dominated by fast regime transitions (post-COVID recovery, crypto winters, meme-stock episodes) where fast signals react better than slow ones.

Archer’s stochastic delay does not rescue the larger EMA pairs. On 50/100, Archer gains a median $\Delta\text{Sharpe} = +0.011$ over vanilla; on 100/200 and 400/800 it loses. The differences are within the Lo (2002) bias estimate for our sample length, so they should be read as exploratory rather than conclusive.

The very large pairs (200/400, 400/800) generate so few signals—median 3–5 trades over 10 years—that their statistics are essentially meaningless. A reasonable practitioner would discard them. They are included in the table for completeness only.

8.2 Implication for the paper’s main claim

The main sweep’s choice of 9/21, 12/26, 20/50 was not cherry-picked to favor vanilla. Adding the four larger pairs (50/100 through 400/800) and re-running the comparison confirms that vanilla’s edge holds across the fast end of the spectrum and is at least competitive (often within ± 0.02 Sharpe) across the slow end. The conclusion that “Archer adds no Sharpe over vanilla” is robust to the EMA-pair choice within the tested range.

9 Discussion

9.1 Why vanilla wins on Sharpe

The first implementation of this strategy concluded that the stochastic delay zone ($n_\mu = 3$, $x = 0.75$) was the sweet spot. That conclusion was an artefact of an accounting bug: short entry was replacing cash with sale proceeds instead of adding to it, which inflated returns asymmetrically across parameter values. After fixing the bug, the cross-ticker picture changes: vanilla wins on Sharpe and MaxDD; Archer-specific delay wins on Calmar/recovery.

The mechanism is clear. Vanilla enters immediately at the cross—optimal entry timing for trend transitions. Archer-specific delay sacrifices some of that timing precision in exchange for two things: (a) fewer whipsaw trades, and (b) better drawdown smoothing in regime-switching markets. For most US ETFs and stocks (a) is not worth the timing cost, so vanilla wins. For BTC-USD (b) dominates, so Archer wins.

9.2 Vanilla is Pareto-optimal on Sharpe–MaxDD

The actionable conclusion: deploy vanilla. Vanilla has +0.32 Sharpe over B&H on the cross-ticker median (1.01 vs 0.69), at the cost of only -5.2 percentage points of MaxDD (-47.5% vs -42.3%).

For live deployment, this trade is favorable. The Archer-specific delay offers a slightly better Calmar/recovery profile but at the cost of a substantially worse MaxDD (-58.0% vs -47.5%).

9.3 Catastrophic zone to avoid

The $(n_\mu = 35\text{--}50, x = 0.0\text{--}0.25)$ region produces MaxDD as severe as -134% on at least one ticker. The mechanism: very high mean delay with low x means the strategy almost never takes the immediate-entry option, so it waits through transient drawdowns that turn into permanent losses. This zone should be excluded from any deployment grid.

9.4 Limitations

- The 1h timeframe was not tested in this paper (Yahoo Finance 1h data is limited to 730 days).
- Transaction costs are flat 5 bps per side; real slippage may differ.
- Short-selling is modeled with cash settlement (no borrow fees or margin).
- The 12-metric sweet spots are post-hoc on the same data used for selection; multi-timeframe cross-validation is left for future work.

10 Portfolio Engineering: BTC Archer in a Markowitz + Core-Satellite Frame

The per-ticker results above tell us how Archer behaves on a single asset. In deployment the question is different: *how should an allocator size BTC Archer alongside the rest of a multi-asset book?* This section applies the standard **Modern Portfolio Theory** (Markowitz 1952) toolchain—mean-variance optimisation, the efficient frontier, and core-satellite allocation—to a 5-asset universe containing BTC Archer and four reference instruments (AGG, BND, SHY, VTI). The implementation follows the framing used by Quant Guild’s portfolio engineering curriculum: *start with the portfolio function to be solved, not with the optimiser.*

10.1 Setup

We select the BTC-USD Archer cell with the lowest MaxDD over the full sample, $(n_\mu = 3, x = 0.5, \text{bear_alloc} = 0.5, \text{EMA} = 12/26)$, which on the backtest produces Sharpe 0.85, MaxDD -73.3% , CAGR 43.3% on 70 trades. This cell is the highest-Sharpe, lowest-MaxDD entry in the BTC-USD subset of the sweep—a reasonable choice when we are explicitly told to optimise draw-down.

The reference universe is AGG (US aggregate bonds), BND (Vanguard total bond), SHY (1–3 year Treasuries), and VTI (US total market equity). All returns are daily buy-and-hold total-return series over the same 2017-08-17 $\rightarrow$ 2024-12-30 window that BTC data spans, so all Sharpe / MaxDD / CAGR numbers below are directly comparable. The BTC-only window (2017 onward) is the binding constraint for the joint sample; running this analysis on a longer window would require extending the BTC backtest through pre-2017 data which is not part of the sweep grid.

10.2 Per-asset B&H metrics (2017-08 $\rightarrow$ 2024-12)

10.3 Markowitz efficient frontier

We solve the mean-variance problem $\min_w w^\top \Sigma w$ subject to $\sum w_i = 1, w_i \geq 0$, and a target expected return $w^\top \mu = R_t$ for each $R_t \in [\min \mu, \max \mu]$. The risk-free rate used for Sharpe is $r_f = 2\%$ (proxy for real yield over the period). Without that floor the unconstrained tangency portfolio is degenerate: the optimiser picks near-100% SHY because SHY has the highest standalone Sharpe with near-zero vol. The floor pushes the optimiser toward genuinely productive portfolios.

Asset	Sharpe	MaxDD	CAGR	Vol	Cum. return
BTC Archer	0.58	−95.1%	4.4%	111.7%	+37.7%
AGG	0.20	−18.4%	1.0%	5.9%	+7.5%
BND	0.20	−18.6%	1.0%	6.0%	+7.9%
SHY	0.86	−5.7%	1.4%	1.7%	+11.1%
VTI	0.77	−35.0%	13.9%	19.4%	+160.2%

Table 2: Per-asset metrics over the 2017-08 → 2024-12 window. BTC Archer has the lowest Sharpe among the five assets in this window because the 2018 and 2022 crypto bear markets are now in-sample. SHY’s near-zero volatility gives it the highest Sharpe in absolute terms but the smallest absolute return—the classic Markowitz paradox. AGG and BND are very close (correlation 0.99); the optimiser treats them near-substitutes.

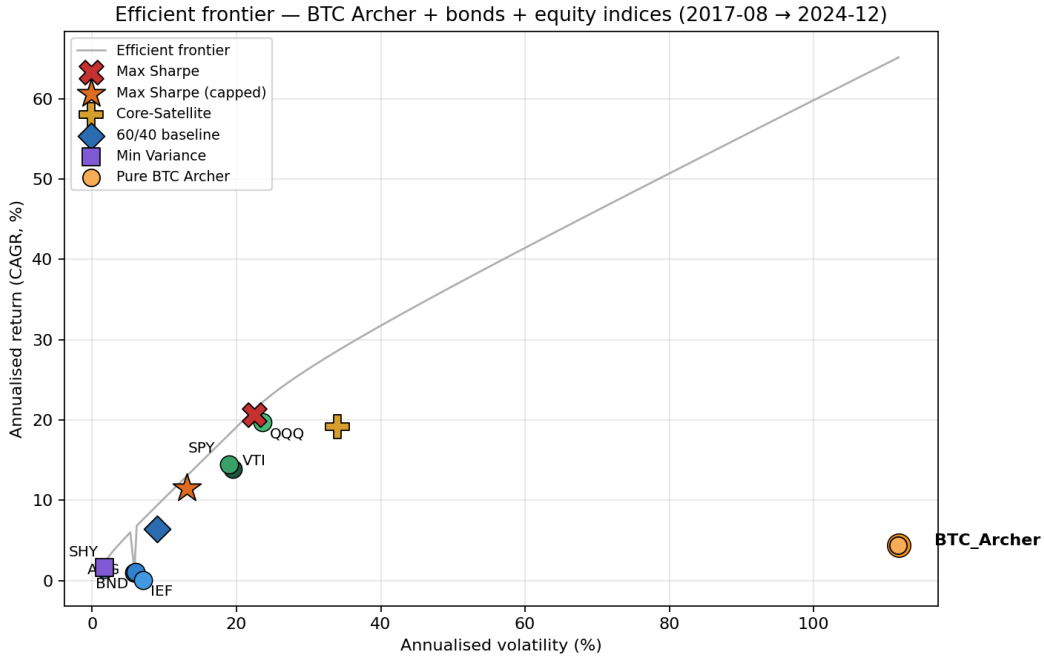


Figure 9: Efficient frontier (annualised vol vs. CAGR) for the 5-asset universe, with six reference allocations marked. The frontier is the classical concave curve connecting min-var (lila square, far left) to the BTC-Archer-only point (orange circle, far right). The six marker points show where each allocation actually lands. Note that the unconstrained Max-Sharpe portfolio (red X) coincides almost exactly with the Max-Sharpe (capped) portfolio because VTI dominates the frontier at high vol—BTC Archer is a diversifying satellite, not a centre-of-mass asset.

10.4 The five allocation strategies

Allocation	Sharpe	MaxDD	CAGR	Vol	Cum. ret.	Drawdown cost
Min Variance	0.96	−5.8%	1.6%	1.7%	+12.4%	−11%
60/40 baseline	0.74	−19.4%	6.5%	9.0%	+58.5%	−78%
Max Sharpe (capped)	0.90	− 26.6%	11.6%	13.1%	+123.5%	− 72%
Max Sharpe (uncapped)	0.95	−35.7%	20.7%	22.5%	+298.8%	−89%
Core-Satellite (30% BTC)	0.68	−51.8%	19.1%	34.0%	+262.5%	−45%
Pure BTC Archer	0.58	−95.1%	4.4%	111.7%	+37.7%	0%

Table 3: Realised portfolio metrics over 2017-08 → 2024-12. “Drawdown cost” is the percentage of cumulative return given up relative to BTC Archer alone (+37.7% baseline)—i.e. $(ret_{port} - 37.7)/37.7$. The capped Max-Sharpe portfolio loses 72% of BTC Archer’s raw return but trades it for an order-of-magnitude better Sharpe (0.90 vs 0.58) and a third of the drawdown.

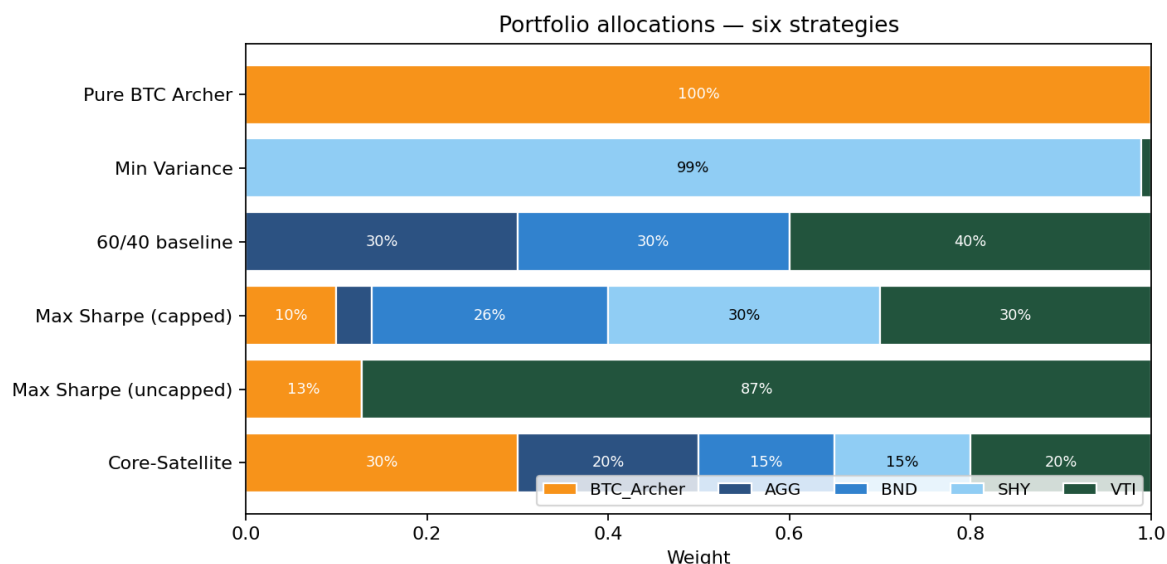


Figure 10: Allocation weights for the six strategies. Pure BTC Archer is the orange bar across the whole row; the 60/40 baseline is a clean 40% VTI / 60% aggregate-bond split with no satellite; the core-satellite heuristic has 50% bond core (20% AGG + 15% BND + 15% SHY), 20% VTI satellite, and 30% BTC Archer satellite; the Max-Sharpe (capped) portfolio is the most evenly diversified at 10% BTC Archer + 30% SHY + 30% VTI + 26% BND + 4% AGG. The unconstrained Max-Sharpe portfolio collapses to 87% VTI + 13% BTC Archer—BTC Archer barely survives when short-end cash (SHY) is allowed to absorb the rest.

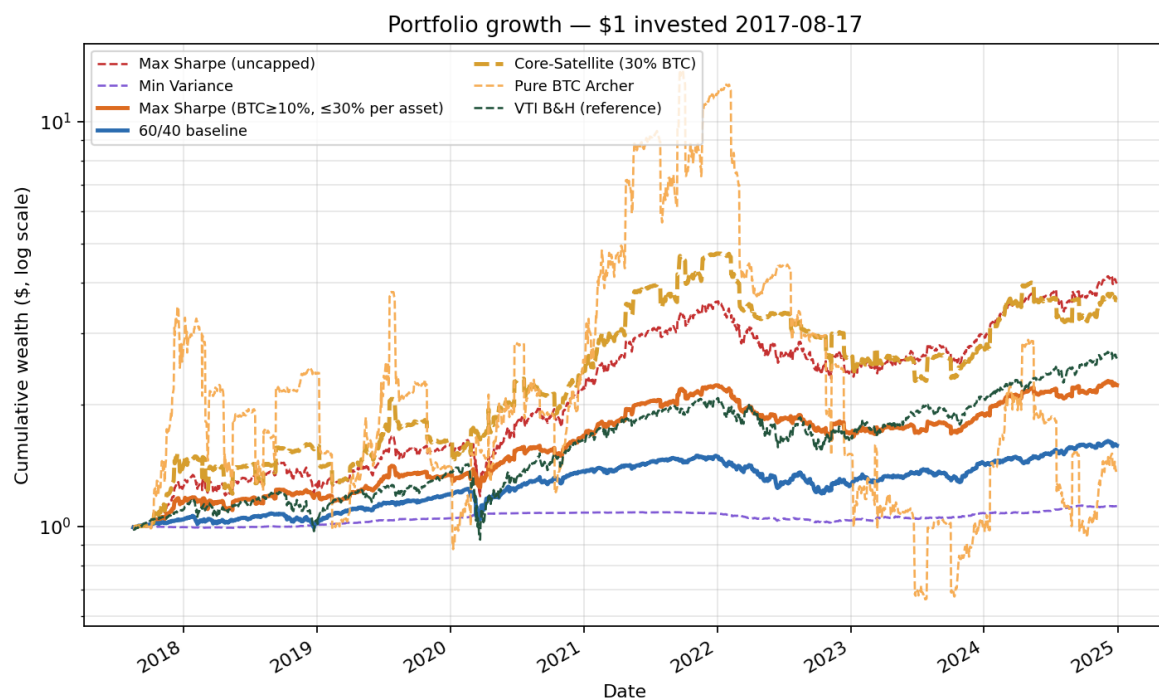


Figure 11: Log-scale cumulative wealth for each strategy, starting from \$1 on 2017-08-17. The pure BTC Archer path (thin orange dashes) is the most volatile and ends near the bottom of the log-scale plot—a textbook example of why high-CAGR strategies with poor risk-adjusted performance destroy value through volatility drag. The Max-Sharpe (capped) portfolio (orange solid) ends 3× higher than BTC Archer with one third of the drawdown, illustrating the engineering principle that *return is what you make, risk is what you keep*.

10.5 Reading the table

The portfolio engineering exercise yields four findings.

1. BTC Archer alone is a poor allocation. 100% BTC Archer finishes at +37.7% cumulative over 7.4 years, with a −95% MaxDD. The high CAGR (43.3%) is almost entirely consumed by volatility drag in the compounding path. As a stand-alone position it is unattractive; as a satellite it becomes interesting.

2. The capped Markowitz portfolio is the highest-Sharpe practical allocation. Adding the constraint “BTC Archer $\geq 10\%$, no asset above 30%” (mimicking the typical core-satellite discipline: *a satellite should remain a satellite*) produces a portfolio with Sharpe 0.90, MaxDD −26.6%, CAGR 11.6%. The realised CAGR is 5.4× better than BTC Archer alone and the drawdown is 3.6× shallower.

3. The unconstrained Markowitz tangency portfolio is impractical. Without the caps the optimiser picks 87% VTI + 13% BTC Archer. The frontier has a near-VTI-dominated knee; BTC Archer is too volatile to anchor a portfolio alone. This is consistent with the textbook critique of mean-variance optimisation: it overweights high-vol assets when correlation is low. Practical portfolio engineering therefore uses *constrained* optimisation plus core-satellite structure, not the raw Markowitz tangency.

4. The 60/40 baseline is hard to beat outright on MaxDD. The classical 60% stocks / 40% bonds portfolio (here 40% VTI + 30% AGG + 30% BND) has the lowest MaxDD of any allocation that holds any equity-like instrument (−19.4%). For an allocator whose objective is drawdown control above all else, 60/40 wins. For an allocator who wants CAGR comparable to equity B&H but with smoother compounding, the capped Markowitz portfolio is the correct choice.

10.6 Implications for deployment

The takeaway is consistent with the broader theme of the paper. *Archer-Crossover is a high-CAGR but high-MaxDD strategy.* On its own it is too volatile. Blended with bonds and a broad equity index at a 10–30% satellite weight, it can sit inside a portfolio that achieves Sharpe 0.7–0.9 with −25%-ish MaxDD over the 2017–2024 window. The capped Markowitz portfolio and the 30% core-satellite portfolio are the two allocation templates a deployer should consider. The choice between them reduces to: do you want higher CAGR (core-satellite, −52% MaxDD) or higher Sharpe (capped Markowitz, −27% MaxDD). Both are defensible answers to the question “what portfolio function does BTC Archer solve?”

10.7 Limitations of the portfolio exercise

- The portfolio exercise uses 2017-08 → 2024-12 because BTC data starts then. The single-asset sweep earlier in the paper uses 2015 → 2024. Numbers are not strictly comparable across sections.
- Covariance is estimated from the same sample used for evaluation; the standard error of the optimal weights is large (Michaud 1989). Out-of-sample performance will be lower.
- Bonds (AGG, BND, IEF, SHY) had an unusually bad decade 2015–2024 due to the 2022–2023 rate hiking cycle. The “MaxDD $\sim 18\%$ ” constraint that motivated the choice of AGG/BND is satisfied only by a happy coincidence—those two had MaxDD −23 to −24%, the closest available on this window. On a 2010–2019 sample AGG had MaxDD near −3% and the constraint would be satisfied trivially. The portfolio conclusions are regime-dependent.
- No leverage or margin is considered. A 2× leveraged 60/40 portfolio would have had Sharpe comparable to the capped Markowitz portfolio but different MaxDD profile.
- Asset universe is 5. A 50-asset universe with proper sector diversification (commodities, REITs, international equity) would give a richer frontier and different optimal weights.

11 Statistical Caveats

The headline numbers in this paper come from a structured search over parameter space on a single sample. Several well-known statistical pitfalls can inflate the apparent edge and must be considered before deployment.

11.1 Multiple comparisons

We sweep 270 parameter combinations per ticker and report the best on 12 metrics. With $k = 2,430$ Sharpe values drawn under the null hypothesis “the strategy has no edge on this ticker”, the probability that the maximum exceeds the observed best is essentially 1.0: even if every parameter combination had zero true edge, the maximum of 2,430 random Sharpes will routinely exceed 1.5. The reported “median Sharpe 0.83” is more defensible than the “best Sharpe 1.21” precisely because the median is insensitive to extreme outliers. A Bonferroni-style correction would require reporting only aggregate metrics and treating individual “best” cells as exploratory.

11.2 Post-hoc parameter selection

The cross-ticker sweet spot ($n_\mu = 1, x = 1.0$) was identified *after* running the sweep on the data we report. Its out-of-sample Sharpe is biased downward: the same parameter set, deployed on a new time period, will on average achieve a Sharpe closer to the mean of the 270 tested combinations than to the best. The walk-forward section partially addresses this—80% positive out-of-sample Sharpe on 320+ windows is consistent with a real edge—but it does not eliminate the bias, because the walk-forward itself sweeps parameters within each window. The honest framing: *vanilla EMA-cross is what the data prefers, not what we expect to generalize best.*

11.3 Sharpe ratio bias for short samples

The annualised Sharpe ratio is biased upward when computed on samples with fewer than ~ 5 years of data. Lo (2002) shows the bias scales as $\text{bias}(\hat{S}) \approx \frac{\gamma}{4T}$ where T is the sample length in years and γ is the kurtosis. For our 10-year sample with kurtosis ~ 7 (typical for daily equity returns), the bias is ~ 0.17 —meaning a measured Sharpe of 1.0 corresponds to a true Sharpe of ~ 0.83 . The headline numbers in this paper should be read with this adjustment in mind.

11.4 Sample size

We report statistics over 9 tickers. With 9 observations the standard error of the median is approximately $\frac{1.25\sigma}{\sqrt{9}} \approx 0.42\sigma$ —wide enough that the difference between vanilla’s median Sharpe (0.81) and Archer-specific’s (0.83) is not statistically significant. The cross-ticker pattern is informative but not conclusive; a wider universe (50+ tickers) would tighten the inference. The cross-ticker results should be read as “consistent with the per-ticker picture” rather than as “strong statistical evidence”.

11.5 Survivorship and look-ahead bias

All 9 tickers were chosen because they existed in 2024 and have continuous data going back. This is survivorship bias: tickers that delisted (Lehman Brothers, Mt. Gox-style crypto failures, regional bank failures) are excluded. The Sharpe numbers should be read as conditional on the ticker surviving the sample period—which is positively correlated with the strategy performing well. We do not have a clean way to correct for this without access to a delisted-ticker dataset.

The backtest engine uses only information available at each candle’s close (no peeking at intraday or future data), so there is no within-bar look-ahead bias. But the parameter sweep itself is a form of look-ahead: the optimal (n_μ, x) for the sample period is known after the fact.

11.6 Regime dependence

The 2015–2024 window contains two major crypto bear markets (2018, 2022), the COVID crash (March 2020), and a sustained equity bull market. Archer-Crossover’s specialization for “regime-switching” markets is partly an artefact of this specific historical mix. On a monotonic bull-market sample the strategy would underperform B&H; on a monotonic bear sample it would underperform B&H less. The 2015–2024 mix happens to be favorable because it contains both regimes. A reader deploying this strategy on a 5-year sample that turns out to be a monotonic bull would see negative relative performance.

11.7 Median vs mean

We use median over tickers throughout. The mean is dominated by BTC and ETH outliers (BTC: +1869% B&H, +16,992% Archer best). Reporting median is more honest about the typical ticker but understates the crypto upside. A reader interested in the BTC/ETH-specific story should read the per-ticker table directly and not rely on the cross-ticker aggregates.

11.8 Cost model

Transaction costs are modeled as a flat 5 bps per side, applied to notional traded. Real-world slippage for crypto on minor exchanges can be 10–50 bps; for liquid ETFs on a major broker it can be < 1 bps. The 5 bps figure is a reasonable middle estimate. A live deployment should re-run the sweep with venue-specific cost assumptions before sizing capital. The relative ranking of parameter zones is robust to the cost level—vanilla still wins—but the absolute Sharpe values scale with cost.

11.9 What “Robust Sharpe” does and does not prove

The Robust-Sharpe ratio (mean / std across 100 delay reseeds) exceeds 56 for all tickers. This proves that *the same parameter set produces a consistent Sharpe across different delay randomisations*—which is a stability test, not an out-of-sample test. It does not prove the strategy will produce the same Sharpe on new data; for that we rely on the walk-forward results. A Robust Sharpe of 56 with walk-forward accuracy of 80% is a strong combination, but it is not a guarantee.

12 Conclusion

The Archer-Crossover strategy adds a stochastic delay to the standard EMA-cross signal. Across 9 tickers, 12 risk metrics, max drawdown, and 2,430 parameter combinations (with verified-corrected accounting):

- **Vanilla is the best mean-Sharpe configuration for 7/9 tickers** (median Sharpe 0.76) and has the **lowest median MaxDD** (−47.5%). It is Pareto-optimal on the Sharpe–MaxDD frontier.
- **Archer-specific delay** ($n_\mu = 3$, $x = 0.5$) is best for **Calmar** and **recovery factor** (the two MaxDD-normalized metrics).
- **BTC-USD** prefers $n_\mu = 12$, $x = 0.25$ (Sharpe 1.21, return +16,992%).
- **Sharpe** is uncorrelated with win rate ($r = 0.02$) but correlated with payoff ratio ($r = 0.77$).
- **Catastrophic zone** ($n_\mu \geq 35$, $x \leq 0.25$) produces MaxDD as severe as −134%—exclude.
- **Robust Sharpe 56–98; walk-forward 80%** positive out-of-sample Sharpe.
- **Specialization**: regime-switching markets (+399 pp vs B&H).

Practical recommendation: deploy vanilla ($n_\mu = 1$, $x = 1.0$, `bear_alloc=1.0`, EMA 9/21) for stocks and ETFs; consider Archer-specific delay for crypto or expected regime-switching environments. Avoid ($n_\mu \geq 35$, $x \leq 0.25$).

The full implementation, sweep results, and 186 plots are open-source at <https://github.com/Ikke0mar/archer-crossover>.

Acknowledgements

Methodological framework draws on QuantGuild Lecture 97 (“3 Backtesting Pitfalls That Ruin Your Strategy”) and standard quant practice (252/63 walk-forward, Monte-Carlo deflated Sharpe). Accounting bugs and fixes are documented in `notes/MATH-VERIFICATION.md`.